Facility status analysis and monitoring

Analyze the current facility status using the Facility Intelligence MCP server. Check the facility overview, energy usage, anomalies, equipment health, and maintenance priorities.

Used facility-intelligence integration, loaded tools ›

Facility Status Analysis

Overall health: DEGRADED. The facility used 11,370.5 kWh today against a baseline of 9,850 kWh, which is 1,520.5 kWh (15.4%) over. The excess comes from Buildings A and C. Building B is running efficiently.

Energy usage

Building	Actual (kWh)	Baseline (kWh)	Variance	Status
Building-A	3,850.5	3,100	+24.2%	Critical excess
Building-B	2,400	2,550	-5.9%	High efficiency
Building-C	5,120	4,200	+21.9%	Critical excess

Equipment faults appear to be driving both overages:

- **Building A:** RTU-3 (compressor short cycling, stuck cooling valve) and RTU-2 (clogged filter).
- **Building C:** Chiller-1 (refrigerant pressure [illegible], fouled condenser) and Cooling-Pump-2 (bearing vibration).

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Maintenance priorities

There are 7 open work orders, estimated at $7,870 in total.

Rank	Work order	Equipment	Issue	Score	Est. cost
1	WO-101	RTU-3	Compressor short cycling, 35% excess peak draw	8.45	$1,250
2	WO-103	Cooling-Pump-2	Bearing vibration above 7.5 mm/s tolerance	7.75	$1,800
3	WO-102	Chiller-1	Condenser fouling, COP down 22%	7.40	$3,400
4	WO-105	Boiler-2	Flame sensor cleaning and recalibration	5.80	$450
5	WO-104	RTU-2	Clogged filter (in progress)	4.60	$250
6	WO-106	AHU-2	Loose drive belt	3.85	$120
7	WO-107	Transformer-1	Annual thermal scan and oil test	3.80	$600

Recommendations

1. **Act on the top three work orders first.** Together they cost about $6,450 and account for most of the energy overage and failure risk.
2. **Treat Cooling-Pump-2 as the most urgent mechanical risk.** Its vibration is more than double the threshold, and a bearing failure could cascade to the chiller loop. It is ranked second by the scoring formula but arguably deserves first-day action.

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Facility status analysis and monitoring

1. **Act on the top three work orders first.** Together they cost about $6,450 and account for most of the energy overage and failure risk.
2. **Treat Cooling-Pump-2 as the most urgent mechanical risk.** Its vibration is more than double the threshold, and a bearing failure could cascade to the chiller loop. It is ranked second by the scoring formula but arguably deserves first-day action.
3. **Finish RTU-2 and the cheap fixes.** The filter ($250), belt ($120) and flame sensor ($450) are low-cost and quick wins for energy and reliability.
4. **Restore Transformer-1 telemetry and do its scan.** It is low-ranked on paper, but it is a critical asset that cannot currently be monitored.
5. **Watch Chiller-1 vibration** (3.9 mm/s) until the condenser is cleaned.

Data inconsistency: the facility status summary reports 5 active anomalies, but the anomaly detector returned 9. The difference is probably that the summary counts only some severities (the 5 non-runtime anomalies would match), but I can't confirm that. It's worth checking before reporting an official figure.

I can also run simulations of corrective actions, such as a setpoint change on RTU-3 or a chiller temperature reset, to estimate energy savings and risk. These are virtual estimates only and don't touch hardware.

Create a Word document

1 minute ago

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don't touch hardware.

Simulate increasing the cooling setpoint of RTU-3 by 1°C. Do not perform any real hardware action. Report the estimated impact, risk, confidence, and safety recommendation.

Simulate action ›

RTU-3 Simulation: +1°C Cooling Setpoint

This was a virtual estimate only. The tool confirms `hardware_modified: false`, and no real action was taken.

Metric	Result
Proposed action	Increase cooling setpoint by 1.0°C
Expected energy impact	-4.2% building cooling load
Estimated power savings	1.63 kW
Estimated risk	**HIGH**
Confidence	82%

Safety recommendation

Do not proceed. The simulation flags caution b[illegible]se RTU-3 has an active critical fault (compressor short cycling and a stuck cooling valve). Its recommendation is to repair the

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manager sign-off before going through the building automation system.

My read

- **The savings are small compared with the problem.** 1.63 kW is minor next to Building A's 750 kWh overage today. RTU-3 alone is causing about 35% excess peak power draw, so repairing it (WO-101, about $1,250) would save far more than any setpoint change.
- **The risk is real.** A setpoint change on a unit with a compressor that is already short cycling and a stuck cooling valve could worsen the cycling or put more stress on the compressor. RTU-3's vibration is also already at 4.1 mm/s, just over threshold.
- **The 82% confidence applies to the estimate, not to safety.** It means the model is fairly sure of the numbers, but the high-risk rating comes from the equipment's fault state.

Suggested next steps

1. Prioritize WO-101 on RTU-3, which is ranked #1 in the maintenance queue.
2. Re-run this simulation after the repair. With the fault cleared, the risk rating should drop and the setpoint change may become a reasonable low-cost efficiency gain.
3. In the meantime, RTU-2's filter replacement (WO-104, already in progress) is a safer way to cut Building A's energy use.

Create a spreadsheet

just now

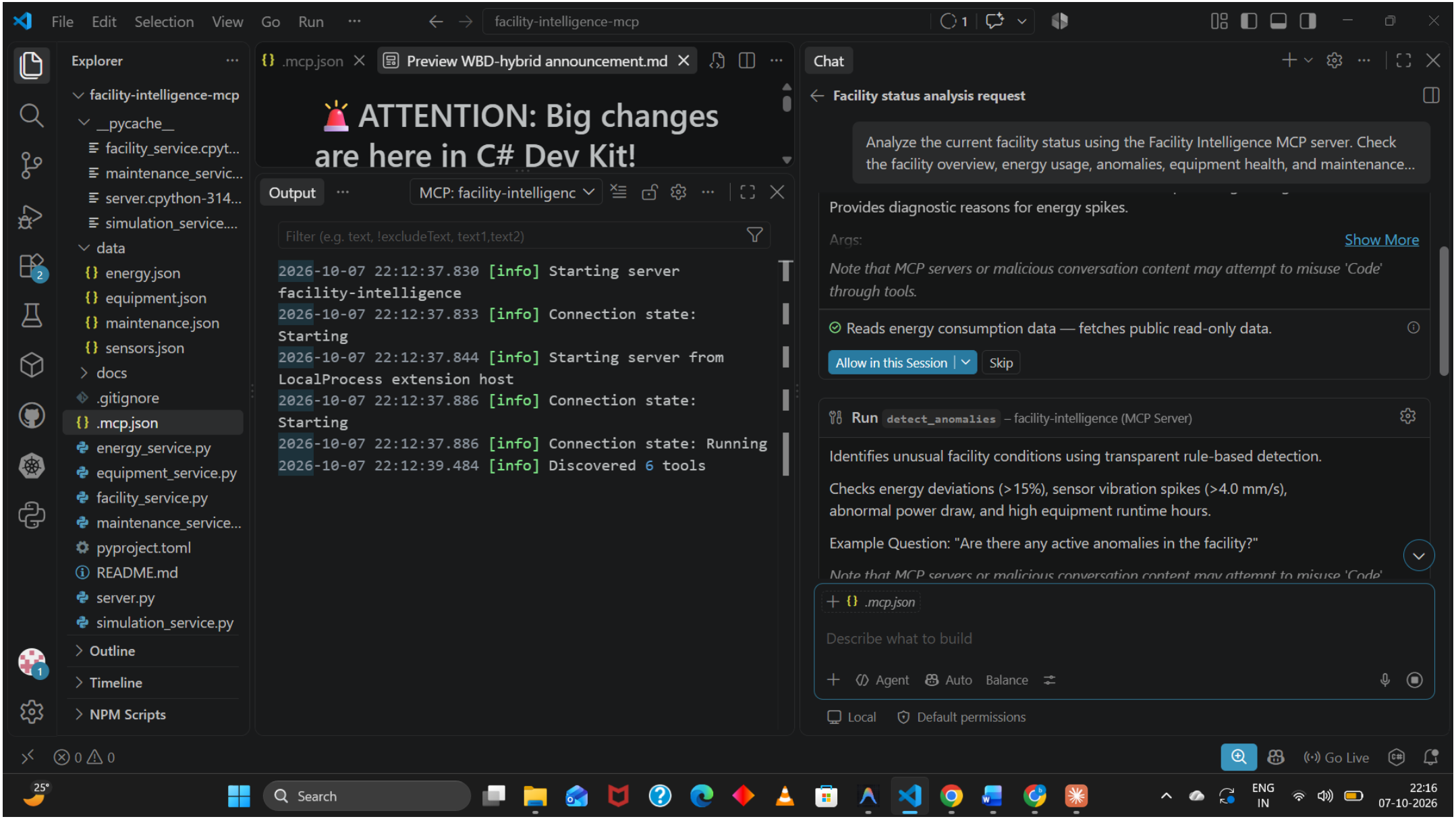
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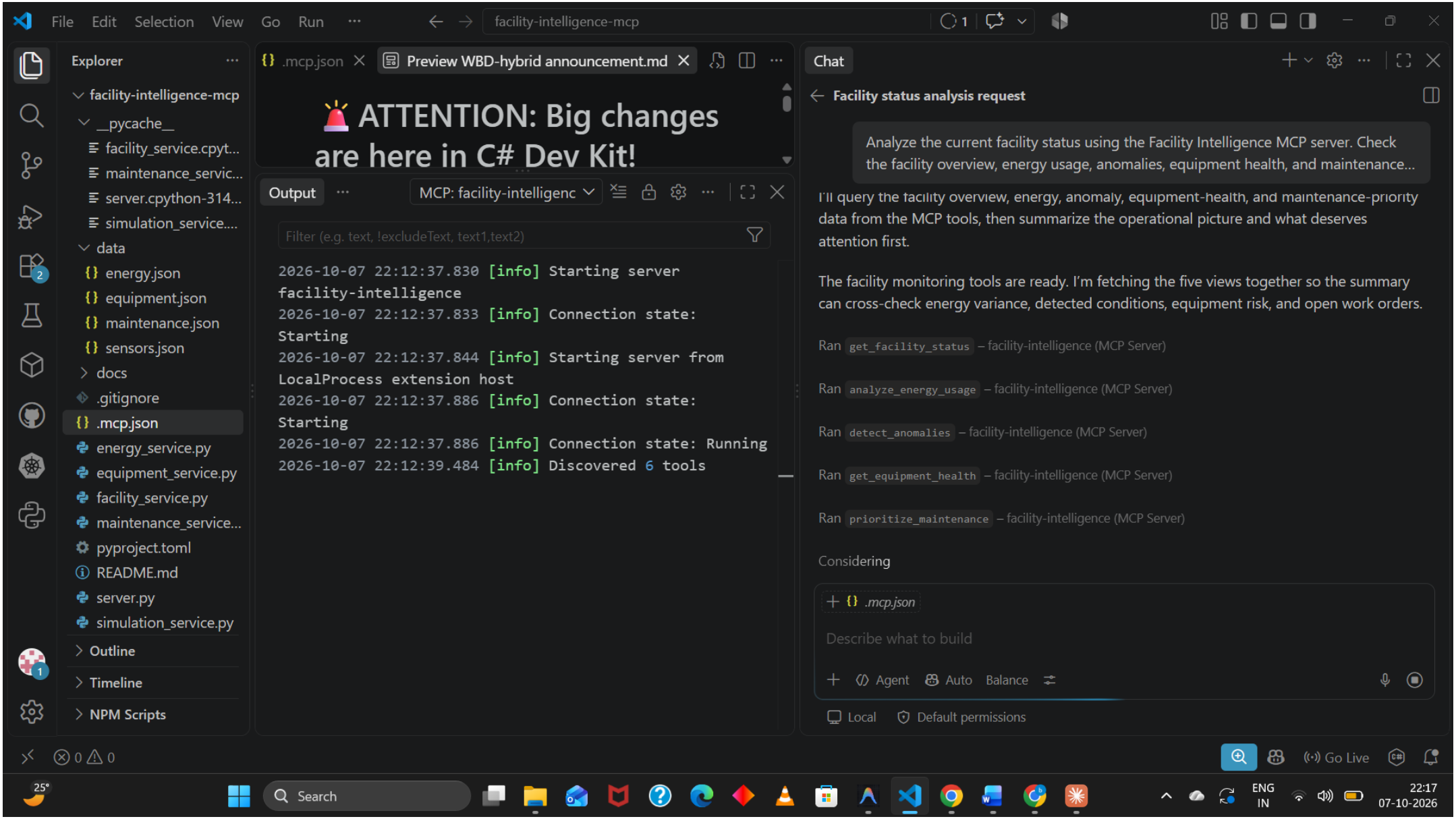
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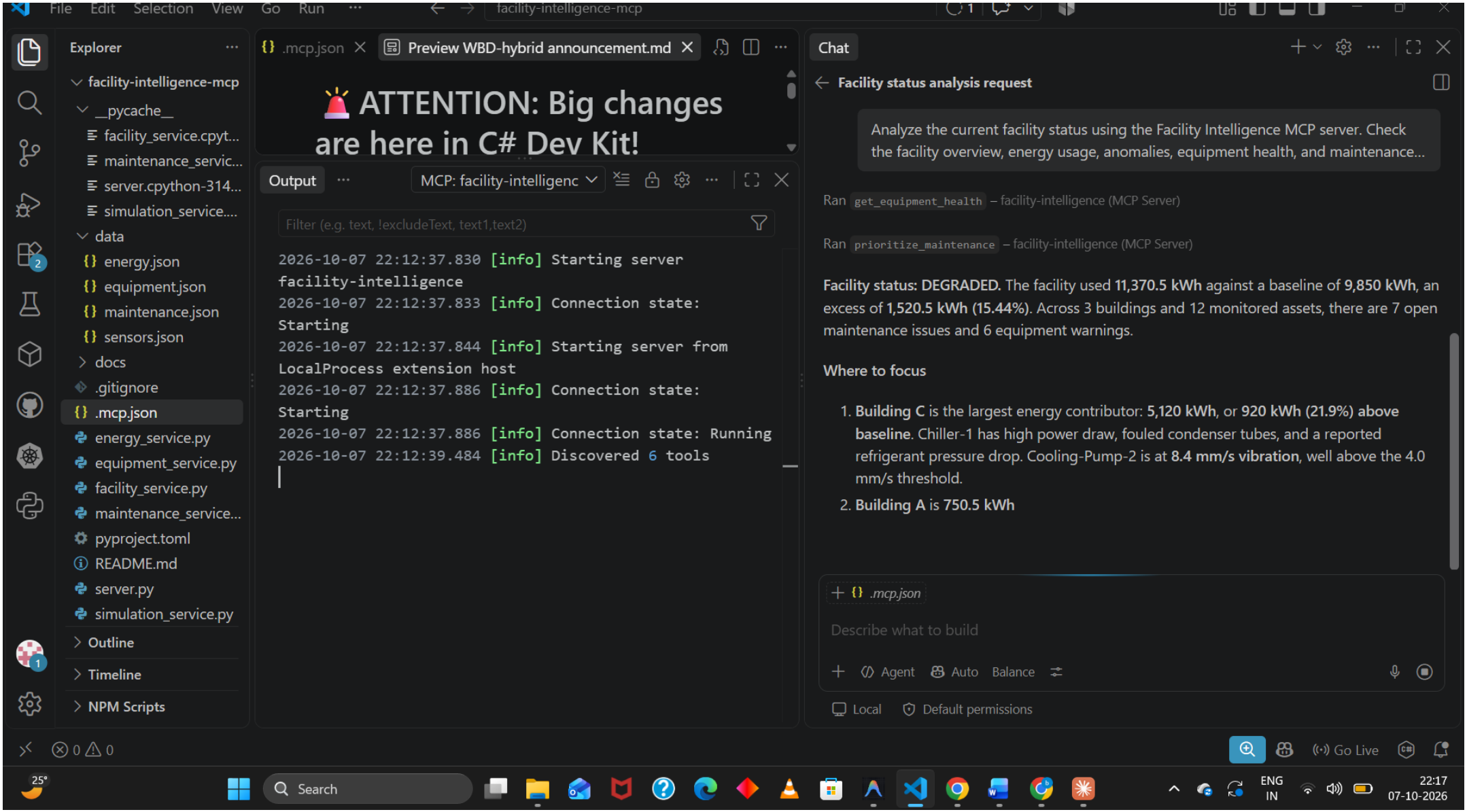
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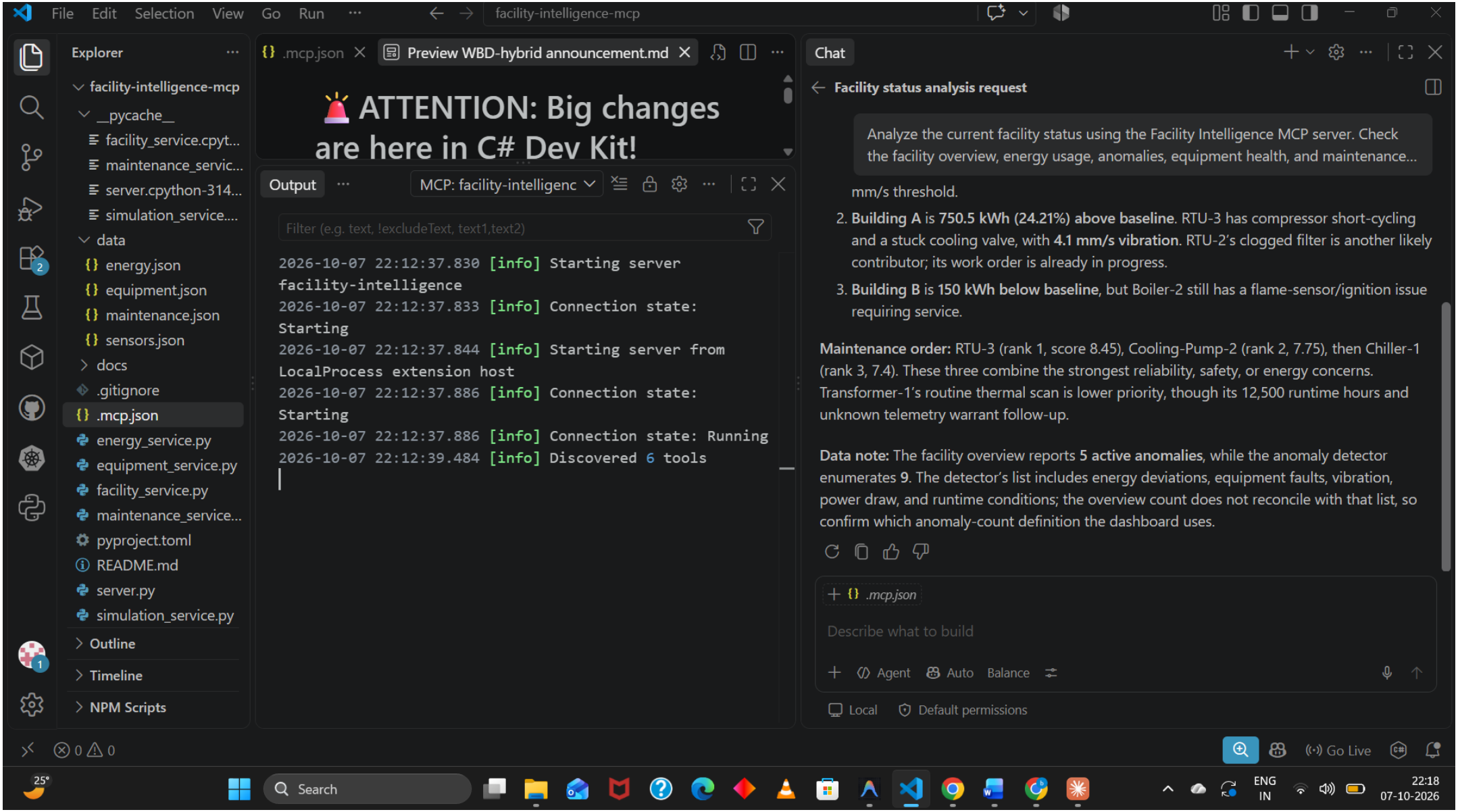
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Search


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facility-intelligence-mcp
__pycache__
facility_service.cpyt...
maintenance_servic...
server.cpython-314...
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.mcp.json
Preview WBD-hybrid announcement.md
ATTENTION: Big changes are here in C# Dev Kit!
Output
MCP: facility-intelligenc
Filter (e.g. text, !excludeText, text1,text2)
2026-10-07 22:12:37.830 [info] Starting server facility-intelligence
2026-10-07 22:12:37.833 [info] Connection state: Starting
2026-10-07 22:12:37.844 [info] Starting server from LocalProcess extension host
2026-10-07 22:12:37.886 [info] Connection state: Starting
2026-10-07 22:12:37.886 [info] Connection state: Running
2026-10-07 22:12:39.484 [info] Discovered 6 tools
Chat
Facility status analysis request
Analyze the current facility status using the Facility Intelligence MCP server. Check the facility overview, energy usage, anomalies, equipment health, and maintenance...
Provides diagnostic reasons for energy spikes.
Args:
Show More
Note that MCP servers or malicious conversation content may attempt to misuse 'Code' through tools.
Reads energy consumption data — fetches public read-only data.
Allow in this Session
Skip
Run detect_anomalies – facility-intelligence (MCP Server)
Identifies unusual facility conditions using transparent rule-based detection.
Checks energy deviations (>15%), sensor vibration spikes (>4.0 mm/s), abnormal power draw, and high equipment runtime hours.
Example Question: "Are there any active anomalies in the facility?"
.mcp.json
Describe what to build
Agent
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Facility status analysis request
Analyze the current facility status using the Facility Intelligence MCP server. Check the facility overview, energy usage, anomalies, equipment health, and maintenance...
I'll query the facility overview, energy, anomaly, equipment-health, and maintenance-priority data from the MCP tools, then summarize the operational picture and what deserves attention first.
The facility monitoring tools are ready. I'm fetching the five views together so the summary can cross-check energy variance, detected conditions, equipment risk, and open work orders.
Ran get_facility_status – facility-intelligence (MCP Server)
Ran analyze_energy_usage – facility-intelligence (MCP Server)
Ran detect_anomalies – facility-intelligence (MCP Server)
Ran get_equipment_health – facility-intelligence (MCP Server)
Ran prioritize_maintenance – facility-intelligence (MCP Server)
Considering
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Describe what to build
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Analyze the current facility status using the Facility Intelligence MCP server. Check the facility overview, energy usage, anomalies, equipment health, and maintenance...
Ran get_equipment_health – facility-intelligence (MCP Server)
Ran prioritize_maintenance – facility-intelligence (MCP Server)
Facility status: DEGRADED. The facility used 11,370.5 kWh against a baseline of 9,850 kWh, an excess of 1,520.5 kWh (15.44%). Across 3 buildings and 12 monitored assets, there are 7 open maintenance issues and 6 equipment warnings.
Where to focus
1. Building C is the largest energy contributor: 5,120 kWh, or 920 kWh (21.9%) above baseline. Chiller-1 has high power draw, fouled condenser tubes, and a reported refrigerant pressure drop. Cooling-Pump-2 is at 8.4 mm/s vibration, well above the 4.0 mm/s threshold.
2. Building A is 750.5 kWh
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Describe what to build
Agent Auto Balance
Local Default permissions
Go Live
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Analyze the current facility status using the Facility Intelligence MCP server. Check the facility overview, energy usage, anomalies, equipment health, and maintenance...
mm/s threshold.
2. Building A is 750.5 kWh (24.21%) above baseline. RTU-3 has compressor short-cycling and a stuck cooling valve, with 4.1 mm/s vibration. RTU-2's clogged filter is another likely contributor; its work order is already in progress.
3. Building B is 150 kWh below baseline, but Boiler-2 still has a flame-sensor/ignition issue requiring service.
Maintenance order: RTU-3 (rank 1, score 8.45), Cooling-Pump-2 (rank 2, 7.75), then Chiller-1 (rank 3, 7.4). These three combine the strongest reliability, safety, or energy concerns. Transformer-1's routine thermal scan is lower priority, though its 12,500 runtime hours and unknown telemetry warrant follow-up.
Data note: The facility overview reports 5 active anomalies, while the anomaly detector enumerates 9. The detector's list includes energy deviations, equipment faults, vibration, power draw, and runtime conditions; the overview count does not reconcile with that list, so confirm which anomaly-count definition the dashboard uses.
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